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United States Patent Application Publication

20250280664

Kind Code

A1

Publication Date

September 04, 2025

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DISPLAY PANEL AND DISPLAY DEVICE

Abstract

The present application provides a display panel and a display device. In the present application, electrical connections between the cathodes of the plurality of sub-pixels are achieved through the plurality of isolation structure. On this basis, the plurality of side walls of each isolation structure are divided into the first side wall and the second side wall. An area enclosed by an outer wall of each isolation structure remains unchanged, the thickness of the first side wall remains unchanged, and the thickness of the second side wall is narrowed or reduces.

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Appl. No.: 19/031447

Filed: January 18, 2025

Foreign Application Priority Data

CN 202410234261.3

Feb. 29, 2024

Publication Classification

Int. Cl.: H10K59/122 (20230101); H10K59/121 (20230101)

U.S. Cl.:

CPC H10K59/122 (20230201); H10K59/121 (20230201);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application claims priority to Chinese Patent Application No. 202410234261.3, entitled “DISPLAY PANEL AND DISPLAY DEVICE”, filed Feb. 29, 2024, which is herein incorporated by reference in its entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to the field of display technologies, and in particular to a display panel and a display device.

BACKGROUND

[0003] At present, organic light-emitting diodes (OLEDs) are prepared by evaporating using fine masks. The fine masks are expensive, so that it is costly to develop new products. Furthermore, a bridging area of openings of the fine mask also limits an effective area of a pixel light-emitting area for evaporation, which does not facilitate improvement of an opening rate.

SUMMARY OF THE DISCLOSURE

[0004] A first technical solution in the present disclosure is to provide a display panel. The display panel includes a pixel definition layer and a plurality of pixel units.

[0005] The pixel definition layer defines a plurality of pixel openings.

[0006] Each of the plurality of pixel units includes a plurality of sub-pixels and a plurality of isolation structures.

[0007] The plurality of sub-pixels are arranged in the plurality of pixel openings one-to-one.

[0008] The plurality of isolation structures protrude from the pixel definition layer and respectively surround the plurality of pixel openings. A plurality of side walls of each of the plurality of isolation structures are defined as a first side wall and a second side wall, respectively. In a direction parallel to the pixel definition layer, a thickness of the first side wall is greater than that of the second side wall. In each of the plurality of isolation structures, the first side wall includes at least a portion of the side walls extending along a first direction and/or at least a portion of the side walls extending along a second direction.

[0009] In each of the plurality of pixel units, the first side wall of at least one isolation structure at least includes an inclined side wall; an extension direction of the inclined side wall intersects with the first direction and the second direction, respectively; and the first direction is perpendicular to the second direction, the extension direction of the inclined side wall, the first direction, and the second direction are located in the same plane.

[0010] A cathode of each of the plurality of sub-pixels is electrically connected to corresponding inclined side wall, so that the cathode of each of the plurality of sub-pixels is electrically connected to a corresponding one of the plurality of isolation structures; and/or, the cathode of each of the plurality of sub-pixels is electrically connected to corresponding side wall extending along the first direction, so that the cathode of each of the plurality of sub-pixels is electrically connected to the corresponding one of the plurality of isolation structures.

[0011] In some embodiments, each of the plurality of isolation structures surrounds a corresponding one of the plurality of pixel openings; there is a one-to-one correspondence between the plurality of side walls of each of the plurality of isolation structures and side edges of each of the plurality of sub-pixels enclosed by the each of the plurality of isolation structures, and the plurality of side walls of the plurality of isolation structures are respectively parallel to the side edges of the plurality of sub-pixels.

[0012] Each of the plurality of sub-pixels includes an anode, a light-emitting layer, and a cathode sequentially stacked; the light-emitting layer and the cathode are sequentially formed on an upper surface of the anode by evaporating an evaporation source; and the first direction is a long side

direction of the evaporation source, and the second direction is a moving direction of the evaporation source.

[0013] In some embodiments, each of the plurality of pixel units includes at least one pixel group, at least one sub-pixel and one of the plurality of isolation structures surrounding corresponding sub-pixel form a pixel group, a shape of each pixel group is a hexagon, the hexagon has an axisymmetric or centrosymmetric shape, and each pixel group includes a side edge extending along the first direction or the second direction.

[0014] Each of the plurality of pixel units includes a plurality of pixel groups, the plurality of pixel groups are disposed side by side and in direct contact with each other along the first direction or the second direction, and a side of one pixel group and a side of an adjacent pixel group close to the side of the pixel group are parallel to each other.

[0015] In some embodiments, the plurality of pixel units are repeated and disposed side by side in the first direction to form a pixel unit row, and a plurality of pixel unit rows are repeated in the second direction.

[0016] Alternatively, the plurality of pixel units are defined as a first pixel unit and a second pixel unit, respectively; the second pixel unit is formed by centrally or axially symmetrizing the first pixel unit; a plurality of first pixel units are repeatedly arranged along the first direction, to form a first pixel unit row; a plurality of second pixel units are repeatedly arranged along the first direction, to form a second pixel unit row; and the first pixel unit row and the second pixel unit row are alternately arranged along the second direction.

[0017] In some embodiments, in the pixel unit, a side of one of the plurality of sub-pixels and a side of an adjacent sub-pixel close to the side of the one of the plurality of sub-pixels are parallel to each other.

[0018] Each pixel group includes an inclined side edge and a side edge extending along the second direction, and an extension direction of the inclined side edge intersects with the first direction and the second direction; each of the plurality of pixel units includes three sub-pixels of different colors, which are defined as a first sub-pixel, a second sub-pixel, and a third sub-pixel, respectively; the first sub-pixel and one of the plurality of isolation structures surrounding the first sub-pixel form a first pixel group; the second sub-pixel, one of the plurality of isolation structures surrounding the second sub-pixel, the third sub-pixel, and one of the plurality of isolation structures surrounding the third sub-pixel form a second pixel group; in the second pixel group, the third sub-pixel and the second sub-pixel share a side wall of the plurality of isolation structures extending along the first direction, and the third sub-pixel is flush with the second sub-pixel in the second direction; and side walls extending along the second direction of two adjacent isolation structures in the first direction overlap in a direction perpendicular to the pixel definition layer.

[0019] In some embodiments, in each of the plurality of isolation structures, the first side wall includes all inclined side walls; in the first direction, two adjacent first side walls are in direct contact with each other and electrically connected to each other.

[0020] In each of the plurality of isolation structures, the first side wall includes the side walls extending along the second direction and a portion of the inclined side walls; in the second direction, every two first side walls of a plurality of first side walls are in direct contact with each other, so as to form a plurality of side wall groups; the plurality of side wall groups are configured to electrically connect two adjacent pixel groups in the second direction; and in the first direction, the plurality of side wall groups are spaced apart from each other.

[0021] In some embodiments, in the second pixel group, the first side wall of each of the plurality of isolation structures further includes a side wall extending along the first direction.

[0022] In some embodiments, each of the plurality of side walls of each of the plurality of isolation structures includes an isolation column and an eave structure disposed on an upper surface of isolation column, and the eave structure covers the isolation column and extends out of the isolation column in the direction parallel to the pixel definition layer; and the eave structure is

conductive or insulating, and the isolation column of the first side wall is conductive and configured for being electrically connected to one of the plurality of sub-pixels that is enclosed by the corresponding one of the plurality of isolation structures.

[0023] In some embodiments, the isolation column of the second side wall is conductive.

[0024] Alternatively, the isolation column of the second side wall is insulating, and the isolation column of the second side wall and the pixel definition layer are formed by patterning the same insulating material.

[0025] In some embodiments, an area of an orthographic projection of the eave structure on the pixel definition layer is greater than an area of an orthographic projection of the isolation column on the pixel definition layer.

[0026] A second technical solution in the present disclosure is to provide a display device including the display panel of any one of above embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0027] In order to more clearly illustrate the technical solutions in some embodiments of the present disclosure, hereinafter, a brief introduction may be given to the accompanying drawings that are used in the description of some embodiments. Obviously, the accompanying drawings in the description below are merely some embodiments of the present disclosure. For those of ordinary skill in the art, other accompanying drawings may be obtained based on these accompanying drawings without any creative efforts.

[0028] FIG. 1 is a structural schematic view illustrating an evaporation source and two limiting plates in an embodiment of related art.

[0029] FIG. 2 is a structural schematic view of a display panel in a first embodiment of the present disclosure.

[0030] FIG. 3 is a structural schematic view of a pixel unit in a first embodiment of the present disclosure.

[0031] FIG. 4 is a cross-sectional structural schematic view of the pixel unit of FIG. 3 in a E-E direction.

[0032] FIG. 5 is a cross-sectional structural schematic view of the pixel unit of FIG. 3 in a F-F direction in an embodiment of the present disclosure.

[0033] FIG. 6 is a cross-sectional structural schematic view of the pixel unit of FIG. 3 in a F-F direction in another embodiment of the present disclosure.

[0034] FIG. 7 is a structural schematic view of the pixel unit in a second embodiment of the present disclosure.

[0035] FIG. 8 is a structural schematic view of the pixel unit in a third embodiment of the present disclosure.

[0036] FIG. 9 is a structural schematic view of the pixel unit in a fourth embodiment of the present disclosure.

[0037] FIG. 10 is a structural schematic view of the pixel unit in a fifth embodiment of the present disclosure.

[0038] FIG. 11 is a structural schematic view of the display panel in a second embodiment of the present disclosure.

[0039] FIG. 12 is a structural schematic view illustrating a first pixel unit and a second pixel unit in a first embodiment of the present disclosure.

[0040] FIG. 13 is a structural schematic view of the display panel in a third embodiment of the present disclosure.

[0041] FIG. 14 is a structural schematic view illustrating the first pixel unit and the second pixel

unit in a second embodiment of the present disclosure.

[0042] FIG. **15** is a structural schematic view of the display panel in a fourth embodiment of the present disclosure.

[0043] FIG. **16** is a structural schematic view of the display panel in a fifth embodiment of the present disclosure.

[0044] FIG. **17** is a structural schematic view of the pixel unit in a sixth embodiment of the present disclosure.

[0045] FIG. **18** is a structural schematic view of the display panel in a sixth embodiment of the present disclosure.

[0046] FIG. **19** is a structural schematic view of the pixel unit in a seventh embodiment of the present disclosure.

[0047] FIG. **20** is a structural schematic view of the display panel in a seventh embodiment of the present disclosure.

[0048] FIG. **21** is a structural schematic view of the pixel unit in an eighth embodiment of the present disclosure.

DETAILED DESCRIPTION

[0049] The technical solutions in some embodiments of the present disclosure may be explained in detail by combining the accompanying drawings.

[0050] In the following description, specific details such as specific system structures, interfaces, technologies, etc. are proposed for the purpose of illustration rather than limitation, so as to fully understand the present disclosure.

[0051] The technical solutions in some embodiments of the present disclosure may be clearly and completely described in conjunction with accompanying drawings in some embodiments of the present disclosure. Obviously, the described embodiments are only a part of the embodiments of the present disclosure, and not all embodiments. Based on the embodiments in the present disclosure, all other embodiments obtained by those of ordinary skill in the art without creative effort are within the scope of the present disclosure.

[0052] The terms “first”, “second”, and “third” in the present disclosure are only configured to describe purposes and cannot be understood as indicating or implying relative importance or implicit indicating the quantity of technical features indicated. Therefore, features limited to “first”, “second”, and “third” may explicitly or implicitly include at least one of these features. In the description of the present disclosure, “multiple” means at least two, such as two, three, etc., unless otherwise expressly and specifically qualified. All directional indications (such as up, down, left, right, front, rear, or the like) in some embodiments of the present disclosure are only configured to explain a relative position relationship between components in a specific posture (as shown in the accompanying drawings), a motion situation between the components in the specific posture (as shown in the accompanying drawings), or the like. If the specific posture is changed, the directional indication is also changed accordingly. In addition, the terms “including”, “comprising”, and “having”, as well as any variations of the terms “including”, “comprising”, and “having”, are intended to cover non-exclusive inclusions. For example, a process, method, system, product, or device that includes a series of operations or units is not limited to the listed operations or units, but optionally includes operations or units that are not listed, or optionally includes other operations or units that are inherent to these processes, methods, products, or devices.

[0053] The reference to “embodiment” in the present disclosure means that, specific features, structures, or characteristics described in conjunction with some embodiments may be included in at least one embodiment of the present disclosure. The phrase appearing in various positions in the specification does not necessarily refer to the same embodiment, nor is it an independent or alternative embodiment that is mutually exclusive with other embodiments. Those of ordinary skill in the art explicitly and implicitly understand that the embodiments described in the present disclosure may be combined with other embodiments.

[0054] An intended purpose of the present disclosure is to provide a display panel and a display device, which solves a problem of how to improve pixel opening rate of the display panel in related art.

[0055] As illustrated in FIG. 1, FIG. 1 is a structural schematic view illustrating an evaporation source and two limiting plates in an embodiment of the related art.

[0056] In actual evaporation process, due to formation of an evaporation cloud in a long side direction of an evaporation source **100**, an evaporation angle of an evaporation material cannot be controlled in the long side direction of the evaporation source **100**. In a moving direction of the evaporation source **100** (i.e., a scanning direction of the evaporation source **100**), an exit angle of the evaporation material may be controlled by using limiting plates **200**, and each of two opposite sides of the evaporation source **100** is provided with one limiting plate **200**. Thus, a side wall of an isolation structure that extends along the long side direction of the evaporation source **100** may be electrically connected to a cathode of a sub-pixel, and a side wall of the isolation structure that extends along the moving direction of the evaporation source **100** cannot be electrically connected to the cathode of the sub-pixel.

[0057] As illustrated in FIGS. 2 to 5, FIG. 2 is a structural schematic view of a display panel in a first embodiment of the present disclosure, FIG. 3 is a structural schematic view of a pixel unit in a first embodiment of the present disclosure, FIG. 4 is a cross-sectional structural schematic view of the pixel unit of FIG. 3 in a E-E direction, and FIG. 5 is a cross-sectional structural schematic view of the pixel unit of FIG. 3 in a F-F direction in an embodiment of the present disclosure.

[0058] Based on the above evaporation technology, the present disclosure provides a display panel. The display panel includes a pixel definition layer **10** and a plurality of pixel units **20**. The pixel definition layer **10** defines a plurality of pixel openings **11**. Each pixel unit **20** includes a plurality of sub-pixels **21** and a plurality of isolation structures **24**. The plurality of sub-pixels **21** are disposed in the plurality of pixel openings **11**, and the plurality of sub-pixels **21** are arranged in the plurality of pixel openings **11** one-to-one. The plurality of isolation structures **24** protrude from the pixel definition layer **10** and respectively surround at least one pixel opening **11**. A plurality of side walls of each isolation structure **24** are defined as a first side wall **241** and a second side wall **242**, respectively. In a direction parallel to the pixel definition layer **10**, a thickness of the first side wall **241** is greater than that of the second side wall **242**. In each isolation structure **24**, the first side wall **241** includes: at least a portion of the side walls extending along a first direction X; and/or at least a portion of the side walls extending along a second direction Y. The first side wall **241** of at least one isolation structure **24** of each pixel unit **20** at least includes an inclined side wall. An extension direction of the inclined side wall intersects with the first direction X and the second direction Y, respectively. The first direction X is perpendicular to the second direction Y, and the extension direction of the inclined side wall, the first direction X, and the second direction Y are located in the same plane. The cathode **213** of each sub-pixel **21** is electrically connected to the corresponding inclined side wall, so that the cathode **213** of each sub-pixel **21** is electrically connected to the corresponding isolation structure **24**; and/or, the cathode **213** of each sub-pixel **21** is electrically connected to the corresponding side wall extending along the first direction X, so that the cathode **213** of the sub-pixel **21** is electrically connected to the corresponding isolation structure **24**.

[0059] In the present disclosure, electrical connections between the cathodes **213** of the plurality of sub-pixels **21** are achieved through the plurality of isolation structure **24**. On this basis, the plurality of side walls of each isolation structure **24** are divided into the first side wall **241** and the second side wall **242**. An area enclosed by an outer wall of each isolation structure **24** remains unchanged, the thickness of the first side wall **241** remains unchanged, and the thickness of the second side wall **242** is narrowed or reduces. Without affecting the electrical connection between each isolation structure **24** and the corresponding cathode **213** of each sub-pixel **21**, the area enclosed by the isolation structure **24** is increased, thereby increasing pixel opening rate of the plurality of sub-pixels **21** and improving resolution of the display panel.

[0060] The pixel definition layer **10** is configured to limit positions of the plurality of sub-pixels **21**. The plurality of pixel openings **11** on the pixel definition layer **10** are spaced apart from each other.

[0061] One sub-pixel **21** is disposed in one pixel opening **11**. One sub-pixel **21** corresponds to one color pixel, and types of the color pixels corresponding to the plurality of sub-pixels **21** are not limited and may be selected based on actual needs.

[0062] Each sub-pixel **21** includes an anode **211**, a light-emitting layer **212**, and a cathode **213** sequentially stacked in a direction from the pixel definition layer **10** to the plurality of isolation structures **24**. The light-emitting layer **212** and the cathode **213** are sequentially formed on an upper surface of the anode **211** through evaporating the evaporation source **100**.

[0063] Each sub-pixel **21** includes an organic light-emitting diode (OLED).

[0064] A shape of each pixel opening **11** determines a shape and a size of the corresponding sub-pixel **21** disposed in each pixel opening **11**. In some embodiments, in the direction parallel to the pixel definition layer **10**, a contour shape of an end of each pixel opening **11** away from the corresponding isolation structure **24** is the shape of the corresponding sub-pixel **21** in each pixel opening **11**. In the direction parallel to the pixel definition layer **10**, the contour shape of the end of each pixel opening **11** away from the corresponding isolation structure **24** is similar to a contour shape of an end of each pixel opening **11** close to the corresponding isolation structure **24**.

[0065] The plurality of isolation structures **24** are disposed on an upper surface of the pixel definition layer **10** and protrudes from the pixel definition layer **10**. The plurality of isolation structures **24** respectively surround at least one pixel opening **11**.

[0066] In response to the plurality of isolation structures **24** respectively surrounding the plurality of pixel openings **11**, the corresponding sub-pixels **21** in the plurality of pixel openings **11** are electrically connected to each other, and each sub-pixel **21** corresponds to one color pixel, so as to avoid pixel crosstalk.

[0067] In response to the plurality of isolation structures **24** surrounding one pixel opening **11**, an area enclosed by inner walls of the plurality of isolation structures **24** determine the shape of the pixel opening **11**. In the direction parallel to the pixel definition layer **10**, the shape of the area enclosed by the inner walls of the plurality of isolation structures **24** is similar to a shape of a cross-section of the pixel opening **11**. That is, the shape of the area enclosed by the inner walls of the plurality of isolation structures **24** is similar to the shape of the sub-pixel **21**. In some embodiments, there is a one-to-one correspondence between the plurality of side walls of each of the plurality of isolation structures **24** and side edges of each of the sub-pixel **21** enclosed by the each of the plurality of isolation structures **24**, and the plurality of side walls of the plurality of isolation structures **24** are respectively parallel to the side edges of the sub-pixel **21**.

[0068] In the present disclosure, one isolation structure **24** surrounding one pixel opening **11** is taken as an example, and distances between the side walls of the isolation structure **24** and the sub-pixel **21** enclosed by the isolation structure **24** are equal.

[0069] In some embodiments, the distances between the side walls of the isolation structure **24** and the sub-pixel **21** enclosed by the isolation structure **24** are not equal.

[0070] The side walls of each isolation structure **24** include an isolation column **243** and an eave structure **244** disposed on an upper surface of isolation column **243**. The eave structure **244** covers the isolation column **243** and extends out of the isolation column **243** in the direction parallel to the pixel definition layer **10**. That is, an orthographic projection of the eave structure **244** on the pixel definition layer **10** covers an orthographic projection of the isolation column **243** on the pixel definition layer **10**. An area of the orthographic projection of the eave structure **244** on the pixel definition layer **10** is greater than an area of the orthographic projection of the isolation column **243** on the pixel definition layer **10**. A structural design of the eave structure **244** allows the evaporation angle to be adjusted through edges of the eave structure **244**, so that the evaporation angle may be adjusted through the eave structure **244** during evaporating the light-emitting layer **212** and the

cathode **213** of the sub-pixel **21**, thereby ensuring that the cathode **213** may cover the light-emitting layer **212** and have a good electrical connection with the corresponding isolation structure **24**.

[0071] The eave structure **244** is conductive or insulating. A material of the eave structure **244** is not limited and may be selected based on actual needs.

[0072] As illustrated in FIG. **3**, the plurality of side walls of each isolation structure **24** are divided into the first side wall **241** and the second side wall **242** that are connected to each other. The thickness of the first side wall **241** is greater than that of the second side wall **242**. The electrical connection between each sub-pixel **21** and the corresponding isolation structure **24** is achieved through the first side wall **241**, and the electrical connection between the plurality of isolation structures **24** is achieved through the first side wall **241**, thereby achieving a full mesh connection between the plurality of sub-pixels **21**, which facilitates uniformity of the cathodes **213**.

[0073] The first side wall **241** of the present disclosure includes at least two side walls.

[0074] The isolation column **243** of the first side wall **241** is conductive and configured for being electrically connected to the corresponding sub-pixel **21** enclosed by the isolation structure **24**. In response to the isolation column **243** being conductive, the isolation column **243** may not only be electrically connected to the corresponding sub-pixel **21**, but also isolate the corresponding sub-pixel **21**.

[0075] In the present disclosure, the eave structure **244** is insulating, and the isolation column **243** of the second side wall **242** is conductive. In response to the isolation column **243** of the second side wall **242** being conductive, entire isolation structures **24** is conductive. Thus, the cathodes **213** of the sub-pixels **21** have better full mesh connectivity, which facilitates the uniformity of the cathodes **213**.

[0076] As illustrated in FIG. **6**, FIG. **6** is a cross-sectional structural schematic view of the pixel unit of FIG. **3** in a F-F direction in another embodiment of the present disclosure.

[0077] In some embodiments, the isolation column **243** of the second side wall **242** is insulating, and the isolation column **243** of the second side wall **242** and the pixel definition layer **10** are formed by patterning the same insulating material. In response to the isolation column **243** of the second side wall **242** being insulating, the isolation column **243** only isolates the sub-pixel **21** and is not electrically connected to the cathode **213** of the sub-pixel **21**. As illustrated in FIG. **6**, the insulating isolation column **243** and the pixel definition layer **10** are formed by patterning the same insulating material, which may simplify preparation process of the isolation column **243**.

[0078] A material of conductive isolation column **243** or a material of insulating isolation column **243** is not limited, and may be selected based on actual needs. In response to the isolation column **243** being insulating, the material of the isolation column **243** may be other insulating materials that are different from the material of the pixel definition layer **10**.

[0079] In the present disclosure, surfaces of all isolation columns **243** away from the pixel definition layer **10** are located in the same plane, so as to simplify the preparation process of the isolation columns **243**.

[0080] The side walls of the isolation structures **24** may also be other structures.

[0081] In each isolation structure **24**, the first side wall **241** includes: at least a portion of the side walls extending along the first direction X; and/or at least a portion of the side walls extending along the second direction Y. The first side wall **241** of at least one isolation structure **24** of each pixel unit **20** at least includes the inclined side wall. That is, each pixel unit **20** must have an isolation structure **24**, and the first side wall **241** of this isolation structure **24** includes the inclined side wall. The sub-pixel **21** is electrically connected to the inclined side wall, so that the sub-pixel **21** is electrically connected to the corresponding isolation structure **24**; and/or, the sub-pixel **21** is electrically connected to the side wall extended along the first direction X, so that the sub-pixel **21** is electrically connected to the corresponding isolation structure **24**.

[0082] The extension direction of the inclined side wall intersects with the first direction X and the second direction Y, respectively. The first direction X is perpendicular to the second direction Y,

and the extension direction of the inclined side wall, the first direction X, and the second direction Y are located in the same plane.

[0083] In some embodiments, the first direction X represents the long side direction of the evaporation source **100**, and the second direction Y represents the moving direction of the evaporation source **100**.

[0084] Based on the evaporation technology in the related art, the present disclosure utilizes the principle that the side wall extending along the first direction X may be electrically connected to the cathode **213** of the sub-pixel **21** enclosed by the isolation structure **24**, and the inclined side wall intersecting with the first direction X and the second direction Y may also be electrically connected to the cathode **213** of the sub-pixel **21** enclosed by the isolation structure **24**. The area enclosed by the outer wall of the isolation structure **24** remains unchanged, the thickness of the first side wall **241** is greater than that of the second side wall **242**, thereby increasing the area enclosed by the inner wall of the isolation structure **24** and improving the pixel opening rate.

[0085] Compared with a design that the thickness of the side wall extending along the first direction X only remains unchanged, the first side walls **241** of the isolation structures **24** in a single pixel unit **20** in the present disclosure may include more types of side walls, and the isolation structures **24** have more arrangement modes.

[0086] Each pixel unit **20** includes at least one pixel group **22**. At least one sub-pixel **21** and one isolation structure **24** surrounding the sub-pixel **21** form one pixel group **22**. A shape of each pixel group **22** is a hexagon. The hexagon has an axisymmetric or centrosymmetric shape. The pixel group **22** includes side edges extending along the first direction X or the second direction Y.

[0087] The pixel group **22** is set as the hexagon and the hexagon is set as the axisymmetric or centrosymmetric shape, so that the pixel units **20** may be arranged compactly, thereby improving the resolution of the display panel.

[0088] In response to each pixel unit **20** including one pixel group **22** (as illustrated FIG. **10**), the pixel unit **20** is the hexagon. In response to the pixel unit **20** including a plurality of pixel groups **22**, the plurality of pixel groups **22** are disposed side by side and in direct contact with each other along the first direction X or the second direction Y, and a side of one pixel group **22** and a side of the adjacent pixel group **22** close to the side of the pixel group **22** are parallel to each other. That is, a first pixel group **22** includes a first side, a second pixel group **22** that is adjacent to the first pixel group **22** includes a second side, the first side of the first pixel group **22** is close to the second pixel group **22**, the second side of the second pixel group **22** is close to the first pixel group **22**, and the first side of the first pixel group **22** and the second side of the second pixel group **22** are parallel to each other. Thus, the pixel units **20** may be arranged compactly, thereby improving the resolution of the display panel.

[0089] In some embodiments, in each pixel unit **20**, a side of one sub-pixel **21** and a side of the adjacent sub-pixel **21** close to the side of the sub-pixel **21** are parallel to each other. That is, one sub-pixel **21** includes a first side, another sub-pixel **21** that is adjacent to the one sub-pixel **21** includes a second side, the first side of the one first sub-pixel **21** is close to the another second sub-pixel **21**, the second side of the another second sub-pixel **21** is close to the one first sub-pixel **21**, and the first side of the one first sub-pixel **21** and the second side of the another second sub-pixel **21** are parallel to each other. Each pixel group **22** includes an inclined side edge and a side edge extending along the second direction Y. An extension direction of the inclined side edge intersects with the first direction X and the second direction Y. Each pixel unit **20** includes three sub-pixels **21** of different colors, which are defined as a first sub-pixel **21A**, a second sub-pixel **21B**, and a third sub-pixel **21C**, respectively. The first sub-pixel **21A** and the isolation structure **24** surrounding the first sub-pixel **21A** form a first pixel group **22A**. The second sub-pixel **21B**, the isolation structure **24** surrounding the second sub-pixel **21B**, the third sub-pixel **21C**, and the isolation structure **24** surrounding the third sub-pixel **21C** form a second pixel group **22B**. In the second pixel group **22B**, the third sub-pixel **21C** and the second sub-pixel **21B** share a side wall of the

isolation structure **24** that extends along the first direction X, and the third sub-pixel **21C** is flush with the second sub-pixel **21B** in the second direction Y. The side walls extending along the second direction Y of two adjacent isolation structures **24** in the first direction X overlap in a direction perpendicular to the pixel definition layer **10**.

[0090] In the second pixel group **22B**, the third sub-pixel **21C** and the second sub-pixel **21B** share the side wall of the isolation structure **24** extending along the first direction X, which means that the side wall extending along the first direction X of the isolation structure **24** that surrounds the third sub-pixel **21C** also serves as the side wall extending along the first direction X of the isolation structure **24** that surrounds the second sub-pixel **21B**. The third sub-pixel **21C** is flush with the second sub-pixel **21B** in the second direction Y, which means that a side edge of the second sub-pixel **21B** away from the first sub-pixel **21A** is on the same line as a side edge of the third sub-pixel **21C** away from the first sub-pixel **21A**, and a side edge of the second sub-pixel **21B** close to the first sub-pixel **21A** is on the same line as a side edge of the third sub-pixel **21C** close to the first sub-pixel **21A**, as illustrated in FIG. 3.

[0091] The pixel group **22** in some embodiments is symmetrically disposed along the first direction X and symmetrically disposed along the second direction Y, and also has a center symmetric shape. The second sub-pixel **21B** and the third sub-pixel **21C** are symmetrically disposed along the second direction Y, as illustrated in FIG. 3.

[0092] As illustrated in FIGS. 2 and 7 to 10, FIG. 7 is a structural schematic view of the pixel unit in a second embodiment of the present disclosure, FIG. 8 is a structural schematic view of the pixel unit in a third embodiment of the present disclosure, FIG. 9 is a structural schematic view of the pixel unit in a fourth embodiment of the present disclosure, and FIG. 10 is a structural schematic view of the pixel unit in a fifth embodiment of the present disclosure.

[0093] In some embodiments, the pixel group **22** may be symmetric only along the first direction X or the second direction Y (as illustrated FIG. 7), or only has the center symmetric shape (as illustrated FIGS. 8 and 9), or symmetric along the first direction X and the second direction Y respectively (as illustrated FIG. 10), which is not limited and may be selected based on actual needs. The sub-pixels **21** of the pixel group **22** may also be asymmetric shapes (as illustrated in FIGS. 8 and 9).

[0094] In some embodiments, the plurality of pixel units **20** are repeated and disposed side by side in the first direction X, to form a pixel unit row **23**. A plurality of pixel unit rows **23** are repeated in the second direction Y, and two adjacent pixel unit rows **23** are disposed in a staggered manner in the first direction X. That is, structures of all pixel units **20** in the display panel are the same.

[0095] As illustrated in FIGS. 11 to 15, FIG. 11 is a structural schematic view of the display panel in a second embodiment of the present disclosure, FIG. 12 is a structural schematic view illustrating a first pixel unit and a second pixel unit in a first embodiment of the present disclosure, FIG. 13 is a structural schematic view of the display panel in a third embodiment of the present disclosure, FIG. 14 is a structural schematic view illustrating the first pixel unit and the second pixel unit in a second embodiment of the present disclosure, and FIG. 15 is a structural schematic view of the display panel in a fourth embodiment of the present disclosure.

[0096] In some embodiments, as illustrated in FIG. 11, odd row pixel unit row **23** may be arranged in a first array, even row pixel unit row **23** may be arranged in a second array, and adjacent odd row pixel unit row **23** and even row pixel unit row **23** are arranged in a staggered manner in the first direction X. In some embodiments, the plurality of pixel units **20** may be defined as a first pixel unit **20A** and a second pixel unit **20B**, respectively. The second pixel unit **20B** (as illustrated in FIGS. 12 and 14) may be formed by centrally or axially symmetrizing the first pixel unit **20A**. A plurality of first pixel units **20A** are repeatedly arranged along the first direction X, to form a first pixel unit row **23A**. A plurality of second pixel units **20B** are repeatedly arranged along the first direction X, to form a second pixel unit row **23B**. The first pixel unit row **23A** and the second pixel unit row **23B** are alternately arranged along the second direction Y.

[0097] In some embodiments, in each isolation structure **24**, the first side wall **241** includes all inclined side walls. In the first direction X, two adjacent first side walls **241** are in direct contact with each other and electrically connected to each other. All sub-pixels **21** are electrically connected to the inclined side walls, so that all sub-pixels **21** are electrically connected to the isolation structures **24**.

[0098] As illustrated in FIGS. **16** and **17**, FIG. **16** is a structural schematic view of the display panel in a fifth embodiment of the present disclosure, and FIG. **17** is a structural schematic view of the pixel unit in a sixth embodiment of the present disclosure.

[0099] In some embodiments, in the second pixel group **22B**, the first side wall **241** of the isolation structure **24** further includes the side wall extending along the first direction X. The first sub-pixel **21A** is electrically connected to the inclined side wall, so that the first sub-pixel **21A** is electrically connected to the isolation structure **24**. The second sub-pixel **21B** and the third sub-pixel **21C** are electrically connected to the side wall extending along the first direction X, so that the second sub-pixel **21B** and the third sub-pixel **21C** are electrically connected to the isolation structure **24**. Thus, the good electrical connection between the sub-pixels **21** and the isolation structures **24** may further enhanced, so as to reduce load on each cathode **213**.

[0100] As illustrated in FIGS. **18** and **19**, FIG. **18** is a structural schematic view of the display panel in a sixth embodiment of the present disclosure, and FIG. **19** is a structural schematic view of the pixel unit in a seventh embodiment of the present disclosure.

[0101] The display panel in sixth embodiment of the present disclosure has a similar structure to the display panel in the first embodiment of the present disclosure, except that in each isolation structure **24**, the first side wall **241** includes the side walls extending along the second direction Y and a portion of the inclined side walls. In the second direction Y, every two first side walls **241** of the plurality of first side walls **241** are in direct contact with each other, so as to form a plurality of side wall groups **245**. The side wall groups **245** are configured to electrically connect two adjacent pixel groups **22** in the second direction Y. In the first direction X, the side wall groups **245** are spaced apart from each other.

[0102] In some embodiments, in each isolation structure **24**, the first side wall **241** includes the side walls extending along the second direction Y and a portion of the inclined side walls. In the second direction Y, every two first side walls **241** of the plurality of first side walls **241** are in direct contact with each other, so as to form the plurality of side wall groups **245**. The side wall groups **245** are configured to electrically connect two adjacent pixel groups **22** in the second direction Y. In the first direction X, the side wall groups **245** are spaced apart from each other. All sub-pixels **21** are electrically connected to the inclined side walls, so that all sub-pixels **21** are electrically connected to the isolation structures **24**.

[0103] The technical solutions in above embodiments may also improve the pixel opening rate.

[0104] As illustrated in FIGS. **20** and **21**, FIG. **20** is a structural schematic view of the display panel in a seventh embodiment of the present disclosure, and FIG. **21** is a structural schematic view of the pixel unit in an eighth embodiment of the present disclosure.

[0105] In some embodiments, in the second pixel group **22B**, the first side wall **241** of the isolation structure **24** further includes the side wall extending along the first direction X. The first sub-pixel **21A** is electrically connected to the inclined side wall, so that the first sub-pixel **21A** is electrically connected to the isolation structure **24**. The second sub-pixel **21B** is electrically connected to the inclined side wall and the side wall extending along the first direction X, so that the second sub-pixel **21B** is electrically connected to the isolation structure **24**, thereby further enhancing the good electrical connection between the sub-pixels **21** and the isolation structures **24**.

[0106] The arrangement of the sub-pixels **21** in the present disclosure includes but is not limited to above arrangement, and layout of the side walls of the first side wall **241** of the present disclosure includes but is not limited to above layout.

[0107] The present disclosure further provides a display device, and the display device includes

any one of above display panels.

[0108] Different from the related art, the effects of the present disclosure are as follows. The present disclosure provides a display panel and a display device. The display panel includes a pixel definition layer and a plurality of pixel units. The pixel definition layer defines a plurality of pixel openings. Each of the plurality of pixel units includes a plurality of sub-pixels and a plurality of isolation structures. The plurality of sub-pixels are arranged in the plurality of pixel openings one-to-one. The plurality of isolation structures protrude from the pixel definition layer and respectively surround the plurality of pixel openings. A plurality of side walls of each of the plurality of isolation structures are defined as a first side wall and a second side wall, respectively. In a direction parallel to the pixel definition layer, a thickness of the first side wall is greater than that of the second side wall. In each of the plurality of isolation structures, the first side wall includes at least a portion of the side walls extending along a first direction, and/or at least a portion of the side walls extending along a second direction. In each of the plurality of pixel units, the first side wall of at least one isolation structure at least includes an inclined side wall; an extension direction of the inclined side wall intersects with the first direction and the second direction, respectively; and the first direction is perpendicular to the second direction; the extension direction of the inclined side wall, the first direction, and the second direction are located in the same plane. A cathode of each of the plurality of sub-pixels is electrically connected to corresponding inclined side wall, so that the cathode of each of the plurality of sub-pixels is electrically connected to a corresponding one of the plurality of isolation structures; and/or, the cathode of each of the plurality of sub-pixels is electrically connected to corresponding side wall extending along the first direction, so that the cathode of each of the plurality of sub-pixels is electrically connected to the corresponding one of the plurality of isolation structures. In the present disclosure, electrical connections between the cathodes of the plurality of sub-pixels are achieved through the plurality of isolation structure. On this basis, the plurality of side walls of each isolation structure are divided into the first side wall and the second side wall. An area enclosed by an outer wall of each isolation structure remains unchanged, the thickness of the first side wall remains unchanged, and the thickness of the second side wall is narrowed or reduces. Without affecting the electrical connection between each isolation structure and the corresponding cathode of each sub-pixel, the area enclosed by the isolation structure is increased, thereby increasing pixel opening rate of the plurality of sub-pixels and improving resolution of the display panel.

[0109] In above embodiments, description of each embodiment has its own emphasis. Parts that are not detailed in one embodiment may refer to relevant descriptions of other embodiments.

[0110] The above descriptions are only some embodiments of the present disclosure, and are not intended to limit the scope of the present disclosure. Any equivalent structure or equivalent flow transformation made by using the contents of the specification and accompanying drawings of the present disclosure, or directly or indirectly applied to other related technical fields, is included in the scope of the patent protection of the present disclosure.

Claims

1. A display panel, comprising: a pixel definition layer defining a plurality of pixel openings; a plurality of pixel units, wherein each of the plurality of pixel units comprises: a plurality of sub-pixels, arranged in the plurality of pixel openings one-to-one; and a plurality of isolation structures, protruding from the pixel definition layer and respectively surrounding the plurality of pixel openings; wherein a plurality of side walls of each of the plurality of isolation structures are defined as a first side wall and a second side wall, respectively; in a direction parallel to the pixel definition layer, a thickness of the first side wall is greater than that of the second side wall; in each of the plurality of isolation structures, the first side wall comprises: at least a portion of the side walls extending along a first direction; and/or, at least a portion of the side walls extending along a

second direction; wherein in each of the plurality of pixel units, the first side wall of at least one isolation structure at least comprises an inclined side wall; an extension direction of the inclined side wall intersects with the first direction and the second direction, respectively; the first direction is perpendicular to the second direction; and the extension direction of the inclined side wall, the first direction, and the second direction are located in the same plane; wherein a cathode of each of the plurality of sub-pixels is electrically connected to corresponding inclined side wall, so that the cathode of each of the plurality of sub-pixels is electrically connected to a corresponding one of the plurality of isolation structures; and/or, the cathode of each of the plurality of sub-pixels is electrically connected to corresponding side wall extending along the first direction, so that the cathode of each of the plurality of sub-pixels is electrically connected to the corresponding one of the plurality of isolation structures.

2. The display panel according to claim 1, wherein each of the plurality of isolation structures surrounds a corresponding one of the plurality of pixel openings; there is a one-to-one correspondence between the plurality of side walls of each of the plurality of isolation structures and side edges of each of the plurality of sub-pixels enclosed by the each of the plurality of isolation structures, and the plurality of side walls of the plurality of isolation structures are respectively parallel to the side edges of the plurality of sub-pixels; each of the plurality of sub-pixels comprises an anode, a light-emitting layer, and a cathode sequentially stacked; the light-emitting layer and the cathode are sequentially formed on an upper surface of the anode by evaporating an evaporation source; and the first direction is a long side direction of the evaporation source, and the second direction is a moving direction of the evaporation source.

3. The display panel according to claim 2, wherein each of the plurality of pixel units comprises at least one pixel group, at least one sub-pixel and one of the plurality of isolation structures surrounding corresponding sub-pixel form a pixel group, a shape of each pixel group is a hexagon, the hexagon has an axisymmetric or centrosymmetric shape, and each pixel group comprises a side edge extending along the first direction or the second direction; and each of the plurality of pixel units comprises a plurality of pixel groups, the plurality of pixel groups are disposed side by side and in direct contact with each other along the first direction or the second direction, and a side of one pixel group and a side of an adjacent pixel group close to the side of the pixel group are parallel to each other.

4. The display panel according to claim 3, wherein: the plurality of pixel units are repeated and disposed side by side in the first direction to form a pixel unit row, and a plurality of the pixel unit rows are repeated in the second direction; or the plurality of pixel units are defined as a first pixel unit and a second pixel unit, respectively; the second pixel unit is formed by centrally or axially symmetrizing the first pixel unit; a plurality of the first pixel units are repeatedly arranged along the first direction, to form a first pixel unit row; a plurality of the second pixel units are repeatedly arranged along the first direction, to form a second pixel unit row; and the first pixel unit row and the second pixel unit row are alternately arranged along the second direction.

5. The display panel according to claim 3, wherein in the pixel unit, a side of one of the plurality of sub-pixels and a side of an adjacent sub-pixel close to the side of the one of the plurality of sub-pixels are parallel to each other; and each pixel group comprises an inclined side edge and a side edge extending along the second direction, and an extension direction of the inclined side edge intersects with the first direction and the second direction; each of the plurality of pixel units comprises three sub-pixels of different colors, which are defined as a first sub-pixel, a second sub-pixel, and a third sub-pixel, respectively; the first sub-pixel and one of the plurality of isolation structures surrounding the first sub-pixel form a first pixel group; the second sub-pixel, one of the plurality of isolation structures surrounding the second sub-pixel, the third sub-pixel, and one of the plurality of isolation structures surrounding the third sub-pixel form a second pixel group; in the second pixel group, the third sub-pixel and the second sub-pixel share a side wall of the plurality of isolation structures extending along the first direction, and the third sub-pixel is flush with the

second sub-pixel in the second direction; and side walls extending along the second direction of two adjacent isolation structures in the first direction overlap in a direction perpendicular to the pixel definition layer.

6. The display panel according to claim 5, wherein: in each of the plurality of isolation structures, the first side wall comprises all inclined side walls; in the first direction, two adjacent first side walls are in direct contact with each other and electrically connected to each other; or in each of the plurality of isolation structures, the first side wall comprises the side walls extending along the second direction and a portion of the inclined side walls; in the second direction, every two first side walls of a plurality of first side walls are in direct contact with each other, so as to form a plurality of side wall groups; the plurality of side wall groups are configured to electrically connect two adjacent pixel groups in the second direction; and in the first direction, the plurality of side wall groups are spaced apart from each other.

7. The display panel according to claim 6, wherein in the second pixel group, the first side wall of each of the plurality of isolation structures further comprises a side wall extending along the first direction.

8. The display panel according to claim 1, wherein each of the plurality of side walls of each of the plurality of isolation structures comprises an isolation column and an eave structure disposed on an upper surface of isolation column, and the eave structure covers the isolation column and extends out of the isolation column in the direction parallel to the pixel definition layer; and the eave structure is conductive or insulating, and the isolation column of the first side wall is conductive and configured for being electrically connected to one of the plurality of sub-pixels that is enclosed by the corresponding one of the plurality of isolation structures.

9. The display panel according to claim 8, wherein the isolation column of the second side wall is conductive; or the isolation column of the second side wall is insulating, and the isolation column of the second side wall and the pixel definition layer are formed by patterning the same insulating material.

10. The display panel according to claim 8, wherein an area of an orthographic projection of the eave structure on the pixel definition layer is greater than an area of an orthographic projection of the isolation column on the pixel definition layer.

11. A display device, comprising: a display panel, comprising: a pixel definition layer defining a plurality of pixel openings; a plurality of pixel units, wherein each of the plurality of pixel units comprises: a plurality of sub-pixels, arranged in the plurality of pixel openings one-to-one; and a plurality of isolation structures, protruding from the pixel definition layer and respectively surrounding the plurality of pixel openings; wherein a plurality of side walls of each of the plurality of isolation structures are defined as a first side wall and a second side wall, respectively; in a direction parallel to the pixel definition layer, a thickness of the first side wall is greater than that of the second side wall; in each of the plurality of isolation structures, the first side wall comprises: at least a portion of the side walls extending along a first direction; and/or, at least a portion of the side walls extending along a second direction; wherein in each of the plurality of pixel units, the first side wall of at least one isolation structure at least comprises an inclined side wall; an extension direction of the inclined side wall intersects with the first direction and the second direction, respectively; the first direction is perpendicular to the second direction; and the extension direction of the inclined side wall, the first direction, and the second direction are located in the same plane; wherein a cathode of each of the plurality of sub-pixels is electrically connected to corresponding inclined side wall, so that the cathode of each of the plurality of sub-pixels is electrically connected to a corresponding one of the plurality of isolation structures; and/or, the cathode of each of the plurality of sub-pixels is electrically connected to corresponding side wall extending along the first direction, so that the cathode of each of the plurality of sub-pixels is electrically connected to the corresponding one of the plurality of isolation structures.

12. The display device according to claim 11, wherein each of the plurality of isolation structures

surrounds a corresponding one of the plurality of pixel openings; there is a one-to-one correspondence between the plurality of side walls of each of the plurality of isolation structures and side edges of each of the plurality of sub-pixels enclosed by the each of the plurality of isolation structures, and the plurality of side walls of the plurality of isolation structures are respectively parallel to the side edges of the plurality of sub-pixels; each of the plurality of sub-pixels comprises an anode, a light-emitting layer, and a cathode sequentially stacked; the light-emitting layer and the cathode are sequentially formed on an upper surface of the anode by evaporating an evaporation source; and the first direction is a long side direction of the evaporation source, and the second direction is a moving direction of the evaporation source.

13. The display device according to claim 12, wherein each of the plurality of pixel units comprises at least one pixel group, at least one sub-pixel and one of the plurality of isolation structures surrounding corresponding sub-pixel form a pixel group, a shape of each pixel group is a hexagon, the hexagon has an axisymmetric or centrosymmetric shape, and each pixel group comprises a side edge extending along the first direction or the second direction; and each of the plurality of pixel units comprises a plurality of pixel groups, the plurality of pixel groups are disposed side by side and in direct contact with each other along the first direction or the second direction, and a side of one pixel group and a side of an adjacent pixel group close to the side of the pixel group are parallel to each other.

14. The display device according to claim 13, wherein: the plurality of pixel units are repeated and disposed side by side in the first direction to form a pixel unit row, and a plurality of the pixel unit rows are repeated in the second direction; or the plurality of pixel units are defined as a first pixel unit and a second pixel unit, respectively; the second pixel unit is formed by centrally or axially symmetrizing the first pixel unit; a plurality of the first pixel units are repeatedly arranged along the first direction, to form a first pixel unit row; a plurality of the second pixel units are repeatedly arranged along the first direction, to form a second pixel unit row; and the first pixel unit row and the second pixel unit row are alternately arranged along the second direction.

15. The display device according to claim 13, wherein in the pixel unit, a side of one of the plurality of sub-pixels and a side of an adjacent sub-pixel close to the side of the one of the plurality of sub-pixels are parallel to each other; and each pixel group comprises an inclined side edge and a side edge extending along the second direction, and an extension direction of the inclined side edge intersects with the first direction and the second direction; each of the plurality of pixel units comprises three sub-pixels of different colors, which are defined as a first sub-pixel, a second sub-pixel, and a third sub-pixel, respectively; the first sub-pixel and one of the plurality of isolation structures surrounding the first sub-pixel form a first pixel group; the second sub-pixel, one of the plurality of isolation structures surrounding the second sub-pixel, the third sub-pixel, and one of the plurality of isolation structures surrounding the third sub-pixel form a second pixel group; in the second pixel group, the third sub-pixel and the second sub-pixel share a side wall of the plurality of isolation structures extending along the first direction, and the third sub-pixel is flush with the second sub-pixel in the second direction; and side walls extending along the second direction of two adjacent isolation structures in the first direction overlap in a direction perpendicular to the pixel definition layer.

16. The display device according to claim 15, wherein: in each of the plurality of isolation structures, the first side wall comprises all inclined side walls; in the first direction, two adjacent first side walls are in direct contact with each other and electrically connected to each other; or in each of the plurality of isolation structures, the first side wall comprises the side walls extending along the second direction and a portion of the inclined side walls; in the second direction, every two first side walls of a plurality of first side walls are in direct contact with each other, so as to form a plurality of side wall groups; the plurality of side wall groups are configured to electrically connect two adjacent pixel groups in the second direction; and in the first direction, the plurality of side wall groups are spaced apart from each other.

17. The display device according to claim 16, wherein in the second pixel group, the first side wall of each of the plurality of isolation structures further comprises a side wall extending along the first direction.

18. The display device according to claim 11, wherein each of the plurality of side walls of each of the plurality of isolation structures comprises an isolation column and an eave structure disposed on an upper surface of isolation column, and the eave structure covers the isolation column and extends out of the isolation column in the direction parallel to the pixel definition layer; and the eave structure is conductive or insulating, and the isolation column of the first side wall is conductive and configured for being electrically connected to one of the plurality of sub-pixels that is enclosed by the corresponding one of the plurality of isolation structures.

19. The display device according to claim 18, wherein the isolation column of the second side wall is conductive; or the isolation column of the second side wall is insulating, and the isolation column of the second side wall and the pixel definition layer are formed by patterning the same insulating material.

20. The display device according to claim 18, wherein an area of an orthographic projection of the eave structure on the pixel definition layer is greater than an area of an orthographic projection of the isolation column on the pixel definition layer.
